

PROFESSIONAL SUMMARY

Software Development Engineer with 4+ years experience building **scalable distributed systems** and **microservices** on **AWS**. Improved **system uptime to 99.9%** and reduced **latency by 30%** in **high-throughput, multi-tenant environments** through strong **system design, API development, and performance optimization**.

EXPERIENCE

Software Engineer | Microsoft

Aug 2024 – Present

- Architected and owned **30+ distributed Java microservices** powering **real-time collaboration** and **version history** for **1M+ daily users**, enabling **seamless co-authoring** across globally scaled, high-availability systems.
- Constructed a **high-throughput backend layer** handling **50K+ requests/sec** using **async workflows, caching**, and optimized **service communication**, reducing **document sync latency by 40%**.
- Engineered **conflict resolution** and **3-way merge algorithms** for concurrent editing, eliminating **data inconsistencies** and improving **collaboration correctness by 30%** across **parallel sessions**.
- Enhanced **system reliability and performance** by implementing **fault-tolerant patterns** (retry, backoff, isolation) and optimizing **data access** (indexing, query tuning), achieving **99.99% availability** and faster **data retrieval at scale**.

Software Development Engineer | DBS Bank

Aug 2022 – Sep 2023

- Designed a **real-time Speech-to-Text and AI summarization platform** integrating **SIP (TCP)** and **RTP (UDP)** pipelines with **OpenAI Whisper**, reducing **manual analysis effort by 60%** across live trading communications.
- Engineered a **scalable session management layer** using **MQTT (Eclipse Paho)** and **WebSockets**, supporting **10K+ concurrent sessions** with **low-latency, reliable communication** across services.
- Modernised **real-time data pipelines** for streaming audio processing using **event-driven architecture** (MQTT, WebSockets), enabling **low-latency ingestion and transformation** at scale, improving **processing throughput by 35%** and ensuring **data reliability**.
- Designed **fault-tolerant scalable data layer** using **Hazelcast (IMap)** and **resilient microservices patterns**, improving **throughput by 35%**, reducing **downtime by 40%**, and cutting **integration time by 50%**.
- Developed **Angular dashboards** and **backend APIs** for **50K+ users**, and automated **CI/CD pipelines** using **Jenkins, Maven, SonarQube, Fortify**, improving **deployment speed by 50%** with strong **code quality and security compliance**.

Associate Software Engineer | DBS Bank India

Jun 2021 – Aug 2022

- Automated a **3-day release process** using **Jenkins** and **Docker CI/CD** with **blue-green deployments** and **rollback**, reducing **deployment time to 4 hours** and enabling **zero-downtime releases**.
- Built **Angular-based banking dashboard** serving **500K+ users** and **millions of requests/month**, driving **API design, schema versioning**, and **secure authorization**, ensuring **zero breaking changes** over **14 months**.
- Owned **production reliability** for **6 services** as **DRI**, creating **dependency maps** and **failure-mode analysis**, reducing **RCA time from days to hours** and driving **resilient system redesigns**.

PROJECTS

Distributed Database Stress Testing Framework | C#, .NET Core, PostgreSQL, Kubernetes

2023

- Built a **stress testing framework** in **C# (.NET Core)** simulating **50K+ concurrent connections** on a **multi-node PostgreSQL cluster**, uncovering **replication consistency bugs** under **split-brain conditions**, preventing potential **data loss**.
- Containerized test environments using **Docker** and **Kubernetes** to mirror production topology, capturing **P50/P95/P99 latency metrics** and identifying **partition imbalance**, reducing **tail latency by 40%** on replica reads.

TECHNICAL SKILLS

Languages & Frontend: Java, Python, C++, SQL, TypeScript, React, Angular

Backend & APIs: Spring Boot, REST APIs, gRPC, Microservices, Distributed Systems

Cloud & Infrastructure: AWS (Lambda, S3, DynamoDB, SNS, SQS), Azure, Docker, Kubernetes

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL Server, Amazon Aurora

Testing & Quality: JUnit, Mockito, Integration Testing, TDD, Automated Testing

Core CS: Data Structures & Algorithms, System Design, Design Patterns

Engineering Practices: Fault Tolerance, Secure Coding, Code Reviews, Agile Development

EDUCATION

University of Dayton

Sep 2023 – Dec 2024

Master of Science in Computer Science

Jawaharlal Nehru Technological University

2017 – 2021

Bachelor of Technology in Computer Science

CERTIFICATIONS

- AWS Certified: AWS Cloud Practitioner 2024